🏫 Rate My Teacher - Congressional App Challenge 🚀

Welcome to the team repository for our K-12 Teacher Review Web App!

Our platform allows students, parents, and alumni to safely share feedback, ratings, and helpful tips about teachers. Our app is **100% anonymous**—we do not collect or store any personal data (no student or teacher names, emails, logins, or passwords). This keeps our project safe, private, and compliant with child privacy laws (COPPA).

🛠️ Step 1: Install Required Tools

Every team member needs to install these three free tools before writing or running code.

1. Python 3: The Brains

Python is the programming language that powers our server and communicates with our database.

- **Download:** Go to python.org/downloads and click the yellow **"Download Python 3.xx"** button.
- **CRITICAL STEP FOR WINDOWS (Do not miss this!):**
  - Open the downloaded installer file.
  - Look at the very bottom of the installer window and check the box that says **"Add python.exe to PATH"**.
  - Click **Install Now**. *(If you forget this checkbox, your terminal will say "Python was not found"!)*
- **How to verify it worked:**
  - Open your terminal (`cmd` on Windows or `Terminal` on Mac).
  - Type: `python --version`
  - If it shows `Python 3.12.x` (or similar), you are ready!

2. Git: The Teleporter

Git lets us download the project code and push our updates to GitHub without overwriting each other's work.

- **Windows:** Download from git-scm.com. Run the installer and click **"Next"** on all default options until finished.
- **Mac:** Open your Terminal (`Cmd + Space`, type `Terminal`, press Enter) and type:

  ```
  git --version
  ```

  If Git is not installed, macOS will display a prompt asking to install "Command Line Developer Tools." Click **Install**.

3. DB Browser for SQLite: The Spyglass

SQLite stores all our data in a single local file. DB Browser is a free visual tool (like Microsoft Excel) that lets you open our database, view the tables, and inspect what's inside.

- **Download:** Go to sqlitebrowser.org/dl and download the standard installer for your operating system (Windows or Mac).
- Follow the on-screen steps to install it.

🗺️ Step 2: The Project Map (Where do files belong?)

To keep our project organized, we use a **3-Tier Layered Architecture**. Think of our app like a **Restaurant**:

```
👫 CUSTOMER (Browser)  <--->   🏃 WAITER (Server)  <--->  🍳 KITCHEN (Database)
   [website/ folder]             [server/ folder]         [database/ folder]
```

📁 Directory Layout

```
rate-my-teacher/
│
├── 📂 database/                # 🍳 The Kitchen (Data & Schema)
│   ├── schema.sql              # Blueprint of all tables (teachers, reviews, votes)
│   └── init_db.py              # Script to build or reset the local database
│
├── 📂 server/                  # 🏃 The Waiter (Logic & Routing)
│   ├── db_gateway.py           # SQL queries and helper functions that talk to SQLite
│   └── dispatcher.py           # Flask web server; routes requests between web & database
│
├── 📂 website/                 # 👫 The Dining Room (User Interface)
│   ├── 📂 templates/           # All visual HTML web page layouts
│   │   ├── index.html          # Main landing page (list of teachers)
│   │   ├── teacher.html        # Teacher profile & review feed
│   │   └── add_teacher.html    # Form to submit a new teacher
│   └── 📂 static/              # Visual styling and client interactivity
│       ├── style.css           # Colors, fonts, margins, and layouts
│       └── app.js              # JavaScript for button clicks (e.g., liking a review)
│
├── .gitignore                  # Tells Git to ignore temporary/database files (*.db)
└── README.md                   # This instruction guide!
```

🧭 File Placement Guide

- **If it's an HTML page that users see:** Put it in `website/templates/`.
- **If it's CSS styling, an image, or client-side JavaScript:** Put it in `website/static/`.
- **If it's Python code that routes web requests or handles URLs:** Put it in `server/dispatcher.py`.
- **If it's Python code that runs SQL queries (`SELECT`, `INSERT`):** Put it in `server/db_gateway.py`.
- **If it's SQL table definitions or database migrations:** Put it in `database/schema.sql`.

🎮 Step 3: Download & Set Up the Project

Follow these quests in your terminal to get the app running on your computer!

Quest 1: Clone the Repository

Open your Terminal or Command Prompt and run:

```
cd Desktop
# (Replace with your team's actual GitHub username and repository link)
git clone https://github.com/YOUR_GITHUB_USERNAME/YOUR_REPOSITORY_NAME.git
cd YOUR_REPOSITORY_NAME
```

Quest 2: Install Flask

Flask is the Python library we use to run our web server. Run:

```
pip install Flask
```

Quest 3: Build the Database

Navigate into the `database` folder and run our initialization script to build your local database file from `schema.sql`:

```
cd database
python init_db.py
cd ..
```

You will see a success message: 🎉 `Success! Database created.` An `app_database.db` file will be created in your `database/` folder!

Quest 4: Launch the Server! 🚀

Navigate into the `server` folder and start the dispatcher:

```
cd server
python dispatcher.py
```

Open your web browser (Chrome, Edge, Safari) and go to: 👉 **`http://127.0.0.1:5000`**

You are now viewing your live, locally running web application!

🕵️ Step 4: How to Inspect Your Database

Want to verify and inspect the data inside your database?

1. Open the **DB Browser for SQLite** program you installed earlier.
2. Click **Open Database** in the top toolbar.
3. Browse to your project folder: `rate-my-teacher/database/app_database.db`.
4. Click on the **Browse Data** tab to view tables like `teachers`, `reviews`, `review_responses`, and `tags`!

🤝 Step 5: Team Rules for GitHub (How to Work Together)

To make sure we never lose work or cause code conflicts:

1. **Pull Before You Code:** Always download the latest updates from your teammates before typing new code:

   ```
   git pull origin main
   ```

2. **Commit with Clear Messages:** When you finish a task, save it with a short note explaining what layer you worked on:

   ```
   git add .
   git commit -m "Front-End: Created the search bar in index.html"
   git push origin main
   ```

3. **Never push `.db` files:** Our `.gitignore` file already prevents `app_database.db` from being uploaded. Each team member has their own local database file generated by `init_db.py`.